

Luan Vucetaj

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EDUCATION

Binghamton University, State University of New York

B.S. in Electrical Engineering, Thomas J. Watson College of Engineering

- Honors: Dean's List (2023); First-Year Research Immersion (FRI) program

Binghamton, NY

Expected May 2027

PROFESSIONAL EXPERIENCE

Rudolph Research Analytical

Electrical Engineering Intern

Hackettstown, NJ

Jul 2026 – Aug 2026

- Diagnosed failure points on 50+ rejected PCBs and restored 25 boards to service for field deployment and in-house projects
- Developed a precise test method for a proposed RTD temperature probe in PCB-mounted and 4-pin connector housings
- Benchmarked probes against 400+ in-service machines, confirming readings within one standard deviation of the mean
- Authored an evidence-backed report that informed management's decision to adopt or reject the new probe
- Identified and corrected an erroneous factory assembly schematic that was adopted into production

Bella's Pizza

Store Manager

Chester, NJ

Jul 2020 – Aug 2026

- Oversaw daily operations including inventory, ordering, and cash handling
- Interviewed, hired, and scheduled a summer staff of 6 employees

PROJECT EXPERIENCE

TALON GS — Portable Rocket Telemetry Ground Station

Team Lead · Senior Design Capstone (SDP128)

Sep 2026 – Present

Binghamton University

- Build a portable ground station that receives, decodes, logs, and visualizes live telemetry from solid-motor rockets
- As sole EE, develop the full electrical scope: SDR-based RF receiver chain, RF link budget, antenna selection, telemetry decoding, and data logging/visualization software
- Coordinate with the project sponsor to define the telemetry link specification and end-to-end integration requirements
- Oversee spending of the \$1,500 state-funded budget on parts and software

BU SIM — Jet Fighter Flight Simulator

Electrical Team Lead

Aug 2026 – Present

Binghamton University

- Lead the electrical sub-team in designing the cockpit control system for a full jet-fighter flight simulator
- Direct the electrical architecture for the throttle, stick, rudder, instrument panel, and side consoles
- Track team progress and assign tasks in line with project timelines
- Coordinate with the mechanical team on control integration and wiring

FPGA-Based Metal Detector & Classifier

Personal Project

Apr 2025 – May 2025

Binghamton University

- Designed a battery-powered Colpitts oscillator-based metal detector interfaced with a Basys 3 FPGA
- Classified unique metal washers by measuring shifts in oscillation frequency
- Implemented frequency-measurement logic in Verilog to distinguish washer types in real time
- Built a live-results display using the FPGA's on-board LEDs and seven-segment display

Autonomous FPGA Robot Navigation System

Team Member

Apr 2025 – May 2025

Binghamton University

- Programmed an FPGA-based robot in embedded C to control motors via PWM using memory-mapped I/O
- Leveraged quadrature-encoder feedback and closed-loop control for drift correction and straight-line accuracy
- Built autonomous navigation using ultrasonic sensors with a finite state machine for wall-following and obstacle avoidance
- Debugged timing issues caused by delays from print statements and debug processes

TECHNICAL SKILLS

Hardware/HDL: FPGA development (Basys 3 / Artix-7), Verilog, FSM design, PCB design, embedded systems

Hardware Skills: PWM motor control, quadrature encoders, ultrasonic sensors, analog sensor interfacing, soldering

Tools & Software: Xilinx Vitis, EasyEDA, Autodesk Fusion (Electronics/Eagle), SolidWorks, MS Project, Excel

Relevant coursework: Digital Systems Design, Embedded Systems, Signals & Systems, Electronics, Circuits, Microprocessors

Languages: C (embedded), Verilog